

ACTIVITY USING POSTGRESQL DATABASE

Aim:

To create and manage a Department–Student database in Postgresql using primary keys, foreign keys, SQL JOIN operations, aggregate functions, UPDATE, DELETE, and LEFT JOIN queries.

SQL Commands and Solutions

1. Create the Department Table (Parent Table)

```
CREATE TABLE Department (  
    Department_ID INTEGER PRIMARY KEY,  
    Department_Name TEXT  
);
```

2. Create the Student Table (Child Table)

```
CREATE TABLE Student (  
    Student_ID INTEGER PRIMARY KEY,  
    Name TEXT,  
    Age INTEGER,  
    Department_ID INTEGER,  
    CGPA REAL,  
    FOREIGN KEY (Department_ID) REFERENCES Department(Department_ID)  
);
```

Sample Data (for testing)

```
INSERT INTO Department (Department_ID, Department_Name) VALUES  
(1, 'CSE'),  
(2, 'Computer Science'),  
(3, 'ECE'),  
(4, 'Mechanical'),  
(5, 'Civil');
```

```
INSERT INTO Student (Student_ID, Name, Age, Department_ID, CGPA) VALUES  
(101, 'Bhavya', 20, 1, 8.7),  
(102, 'Anu', 21, 1, 9.1),  
(103, 'Ravi', 20, 2, 8.5),  
(104, 'Kiran', 22, 3, 7.9),  
(105, 'Sita', 21, 4, 8.2);
```

3. Display students from CSE department

```
SELECT s.*  
FROM Student s  
JOIN Department d ON s.Department_ID = d.Department_ID  
WHERE d.Department_Name = 'CSE';
```

4. Count students in each department

```
SELECT d.Department_Name, COUNT(s.Student_ID) AS Student_Count  
FROM Department d  
LEFT JOIN Student s ON d.Department_ID = s.Department_ID  
GROUP BY d.Department_ID, d.Department_Name;
```

5. Update the foreign key (Example: change the department of a student)

```
UPDATE Student  
SET Department_ID = 2  
WHERE Student_ID = 101;
```

6. Delete a department

```
DELETE FROM Department  
WHERE Department_ID = 5;
```

7. Display each student's name with the corresponding department name using SQL JOIN

```
SELECT s.Name AS Student_Name, d.Department_Name  
FROM Student s  
JOIN Department d ON s.Department_ID = d.Department_ID;
```

8. Display the number of students in each department

```
SELECT d.Department_Name, COUNT(s.Student_ID) AS Number_of_Students  
FROM Department d  
LEFT JOIN Student s ON d.Department_ID = s.Department_ID  
GROUP BY d.Department_ID, d.Department_Name;
```

9. Display the names of students belonging to the Computer Science department

```
SELECT s.Name  
FROM Student s  
JOIN Department d ON s.Department_ID = d.Department_ID  
WHERE d.Department_Name = 'Computer Science';
```

10. Update the department of a student by changing the foreign key value and display the updated record

```
UPDATE Student  
SET Department_ID = 3  
WHERE Student_ID = 103;
```

```
SELECT s.Student_ID, s.Name, s.Age, s.Department_ID, s.CGPA, d.Department_Name  
FROM Student s  
JOIN Department d ON s.Department_ID = d.Department_ID  
WHERE s.Student_ID = 103;
```

11. Display departments that currently have no students assigned using an appropriate SQL JOIN

```
SELECT d.Department_ID, d.Department_Name  
FROM Department d  
LEFT JOIN Student s ON d.Department_ID = s.Department_ID  
WHERE s.Student_ID IS NULL;
```